Bihar Engineering University, Patna

End Semester Examination - 2023

Course: B.Tech. Semester-VI Time: 03 Hours

Code: 101617 Subject: Concrete Materials Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper

(iii) Attempt FIVE questions in all

(iv) Question No I is compulsory

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Q.1 Choose/write the correct answer of the following (any seven Question only):

(a) High performance concrete is _____ as compared to normal strength concrete

(i) Less brittle

(ii) Brittle

(iii) More brittle

(iv) Highly ductile

(b) Lightweight concrete has all the following beneficial characteristic except

(i) High thermal insulation

(ii) High sound insulation

(iii) Excellent fire resistance

(iv) Reduced dry shrinkage

(c) Insufficient quantity of water

(i) Makes the concrete mix harsh

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(ii) Makes the concrete mix as workable

(iii) Causes segregation in concrete

(iv) causes bleeding in concrete

(d) The factor which effects the design of concrete mix is

(i) Fineness modulus

(ii) Water cement ratio

(ii) Slump

(iv) All of these

(e) When vibrators are used for compaction, the consistency of concrete depends upon the

(i) Type of mix

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(ii) Placing conditions

(ii) Efficiency of vibrator

(iv) All of these

(f) For continuous transportation of concrete, the method used is

(i) Transport of concrete by pans

(ii) Transport of concrete by wheel barrows

(iii) Transport of concrete by belt conveyors

(iv) Transport of concrete by pumps

(g) What is meant by a crack arrester in concrete?

(h) What is meant by bleeding of concrete?

(i) Describe M-sand

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(j) Describe Nominal mix and design mix

Q. 2 (a) What are the advantages and disadvantages of re-vibration of concrete?

(b) What particular precautions would you take when concreting:

(i) In cold weather

(ii) In hot weather

Q.3 (a) What are the effects of hot weather on

(i) Fresh concrete

(ii) Hardened concrete

(b) What are the disadvantages of membrane curing compared with water curing?

Q.4 (a) What is the effect of the rate of loading on strength of concrete?

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(b) Describe the role of aggregate in creep of concrete

Q. 5 (a) Explain the role of fly ash in minimizing the alkali-silica reaction.

(b) What are the advantages and disadvantages of the pull-out test?

Q.6 (a) Differentiate light weight and normal weight aggregate.

(b) How does light weight aggregate affect the corrosion of reinforcement?

Q.7. List the laboratory test conducted on fresh concrete as per IS code Explain any four in detail. https://www.akubihar.com

Q.8 Explain briefly the following non-destructive testing of concrete

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(i) Rebound Hammer method

(ii) Ultrasonic pulse velocity testing method

Q.9 Write short notes on any four of the following

(a) Self-compacting concrete.

(b) Recycled concrete aggregate

(c) Fibre reinforced concrete

(d) Roller compacted concrete

(e) Ferrocement

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(f) M-sand

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